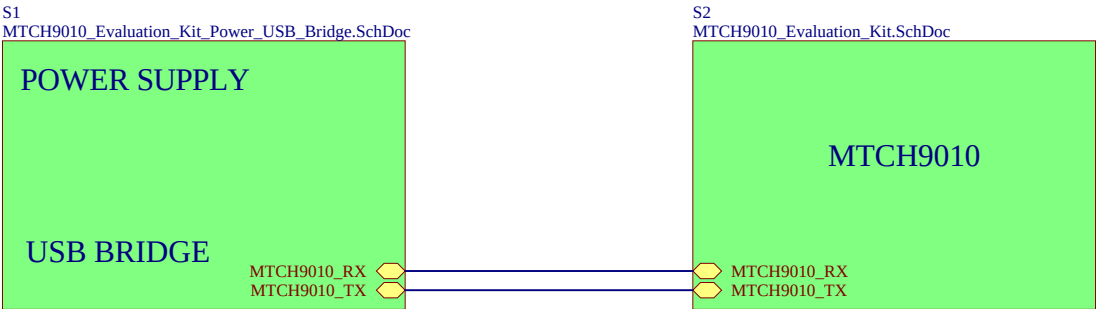


MTCH9010 Evaluation Kit



SCR1
Phillips Screw 1/4"

SCR2
Phillips Screw 1/4"

SCR3
Phillips Screw 1/4"

SCR4
Phillips Screw 1/4"

STANDOFF1
1/4" Standoff Nylon

STANDOFF2
1/4" Standoff Nylon

STANDOFF3
1/4" Standoff Nylon

STANDOFF4
1/4" Standoff Nylon

Test1
\$ >_

TEST, MTCH9010 Evaluation Kit

LABEL1

[ASSY# / REV]
SN: [SERIAL]
[DATE yyyy.mm.dd]

PCBA LABEL 18X6mm

S3
MTCH9010_Evaluation_Kit_Revision_History.SchDoc

Text in Silkscreen on PCB

Project Owner:
SLT

PCB Layout Contact:
SLT

PartNumber:
EV24U22A

Sheet Title
Top Level

Size
A3

Project Title
MTCH9010 Evaluation Kit

Variant: Default Assembly

Rev: 5

Date: 2025-04-14

Rev: 5

Sheet 1 of 4

File: MTCH9010_Evaluation_Kit_TopLevel.SchDoc

Microchip

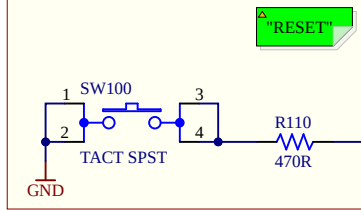
Designed with
Altium

Altium.com

CNANO Template Revision: 2.8

MTCH9010 Evaluation Kit

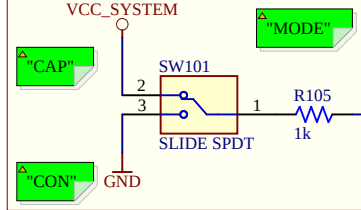
RESET BUTTON



RESET

Use the reset button to latch new board configuration.

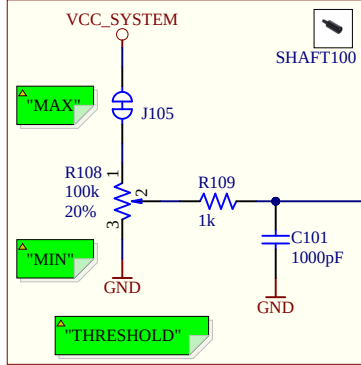
OPERATION MODE SELECTION



OPERATION_MODE

Select between capacitive or conductive operation. Pin setting can be overridden using Enhanced Configuration.

LIQUID DETECTION THRESHOLD

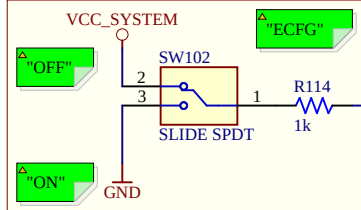


THRESHOLD

Detection threshold: lowest at MIN, highest at MAX. Threshold value can be overridden using Enhanced Configuration.

Detection threshold maximum can be adjusted by soldering a resistor to the cut-strap J105.

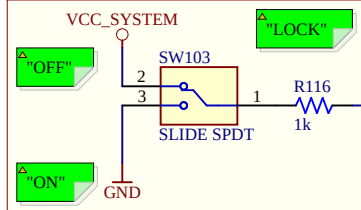
ENHANCED CONFIG ENABLE



ECFG_ENABLE

Enable or disable Enhanced Configuration Mode.

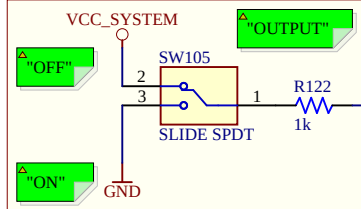
SYSTEM LOCK



SYSTEM_LOCK

Save configuration to NVM; requires Enhanced Configuration Mode.

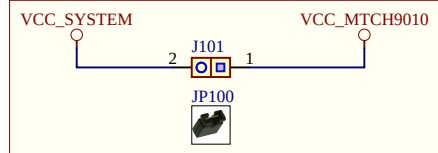
EXTENDED OUTPUT ENABLE



OUTPUT_ENABLE

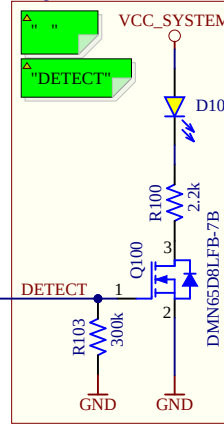
Enable or disable Extended Output. Pin setting can be overridden using Enhanced Configuration.

MTCH9010 CURRENT MEASUREMENT

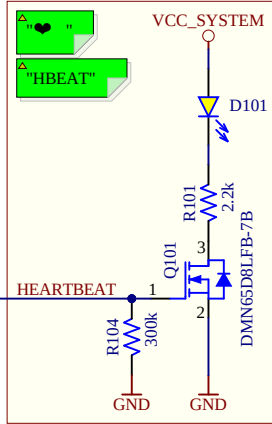


To measure the current consumption of the MTCH9010, remove JP100 and connect an ammeter to J101.

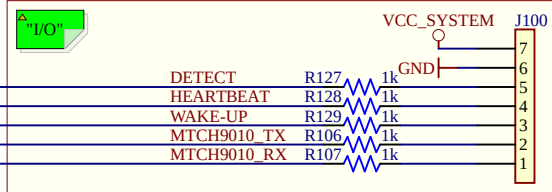
LIQUID DETECT LED



HEARTBEAT LED

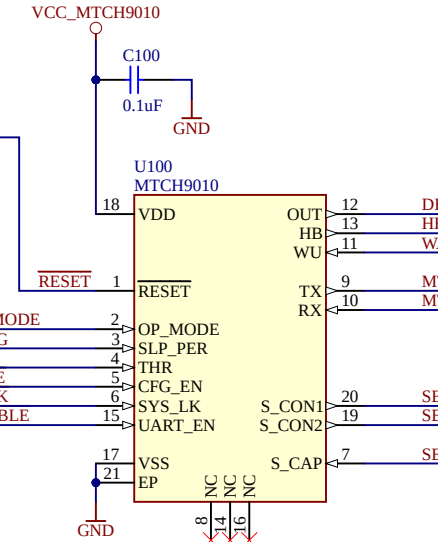


EXTERNAL CONTROL INTERFACE



The J100-7 pin can connect external circuitry to VCC_SYSTEM or receive external power. It is labeled "VDD" in silkscreen.

CAUTION: Power pins on J100 do not have polarity protection. Remove jumper from J201 when supplying external power.



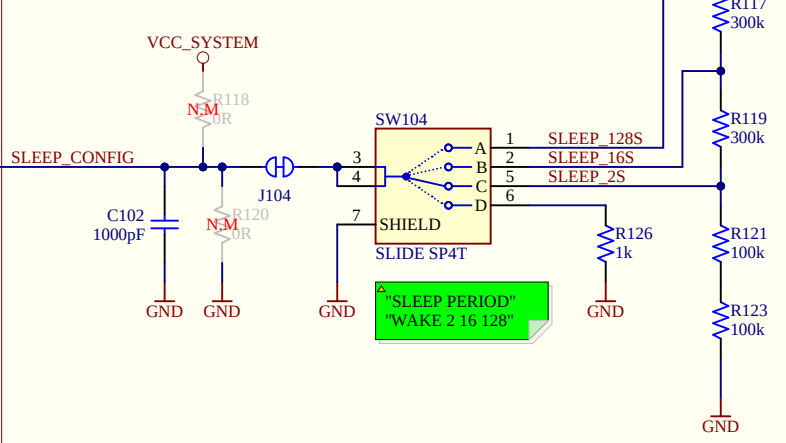
SLEEP PERIOD CONFIGURATION

The slide switch allows selecting a preset sleep period or waking the MTCH9010 by an external WAKE-UP signal.

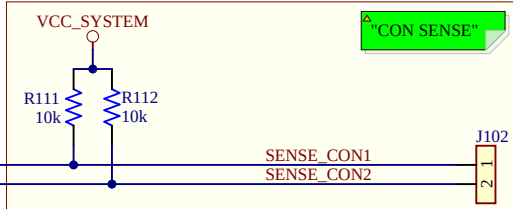
Preset sleep intervals: 2s, 16s, 128s

User-configurable sleep: Cut J104 and add suitable resistors to R118 and R120

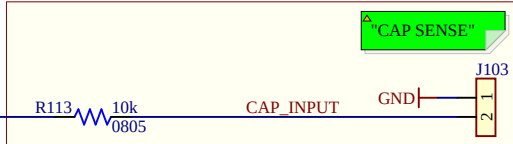
Sleep settings can be overridden using Enhanced Configuration.



CONDUCTIVE SENSOR HEADER

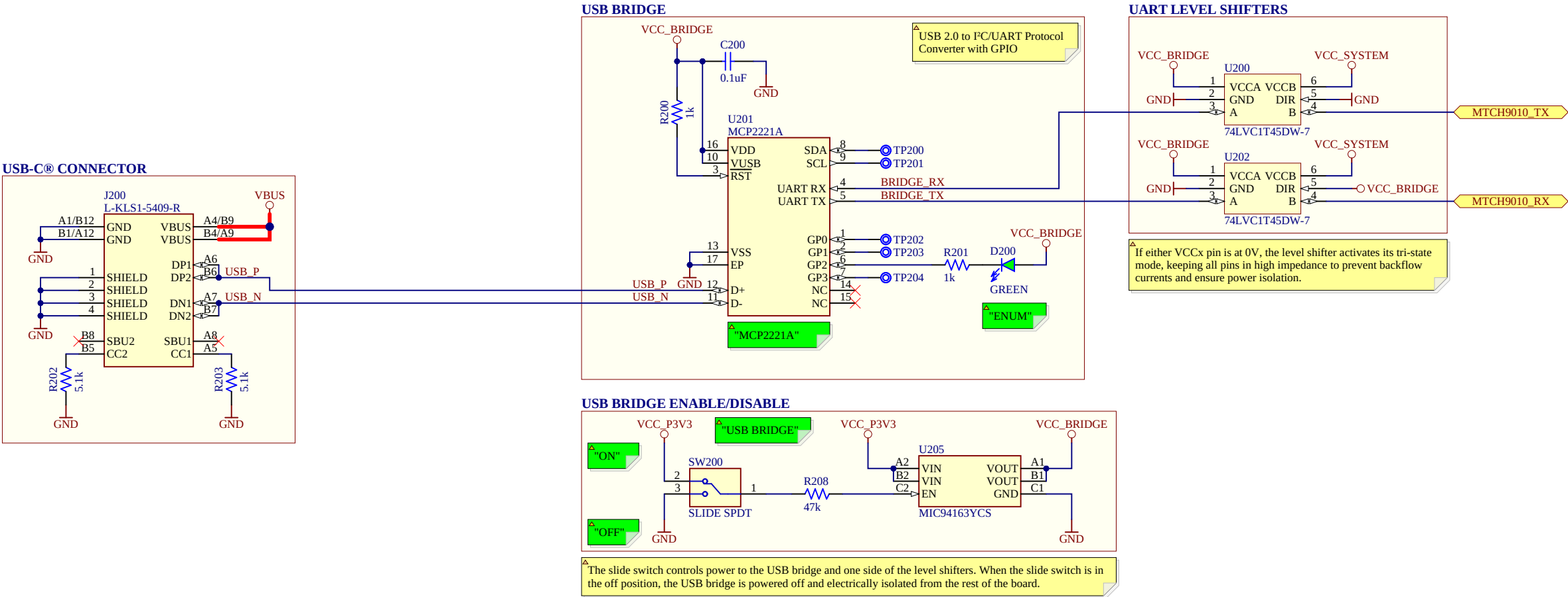


CAPACITIVE SENSOR HEADER

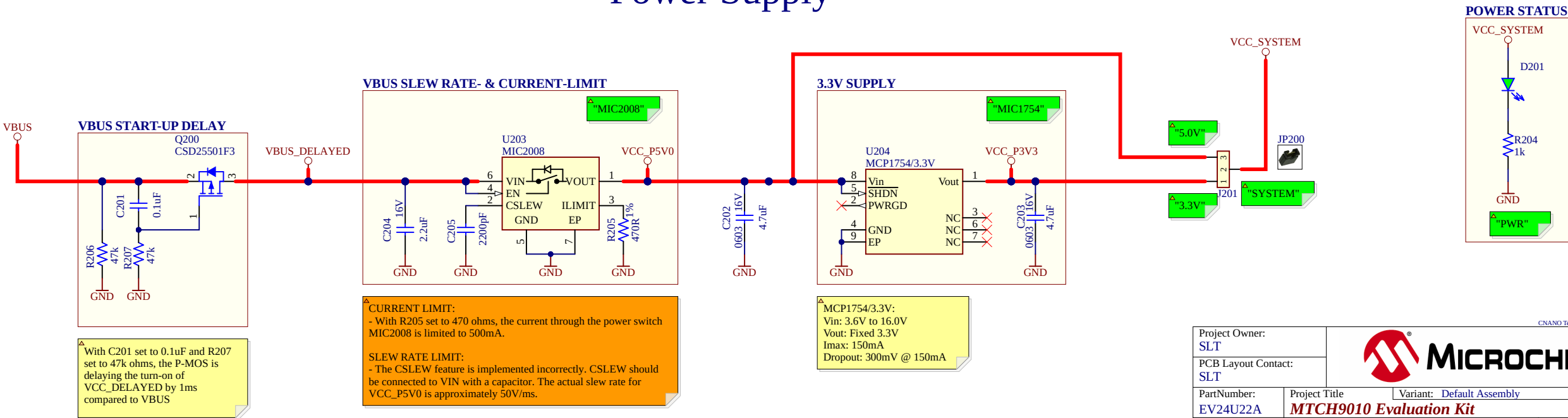


Project Owner: SLT		 MICROCHIP			
PCB Layout Contact: SLT					
PartNumber: EV24U22A		Project Title MTCH9010 Evaluation Kit		Variant: Default Assembly	
Sheet Title MTCH9010 Evaluation Kit					
Size A3	SCH #: 02-01228	Rev: 5	Date: 2025-04-14		Designed with  Altium.com
	PCB #: 04-12292	Rev: 5	Sheet 2 of 4		
File: MTCH9010_Evaluation_Kit.SchDoc					

USB Bridge



Power Supply



Revision History

PCB Assembly Rev 4:

Design Changes:

Initial Design

PCB:

PCB revision 4

PCB Assembly Rev 5:

Design Changes:

- Added capacitor and series resistor to reset circuit

- Changed Enhanced configuration switch connections

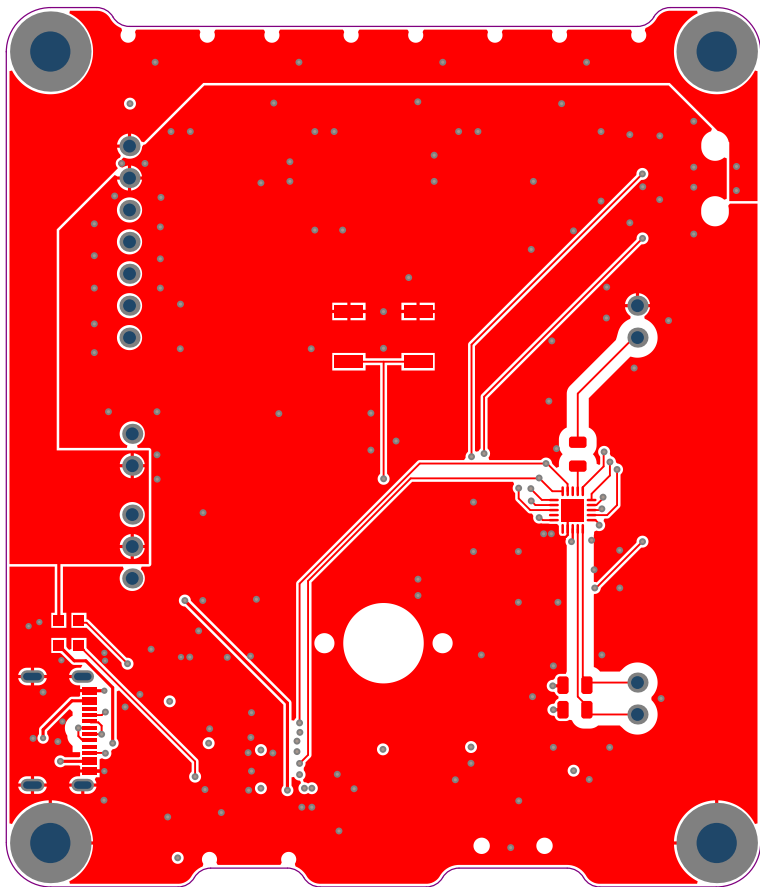
- Changed detection threshold potentiometer connections

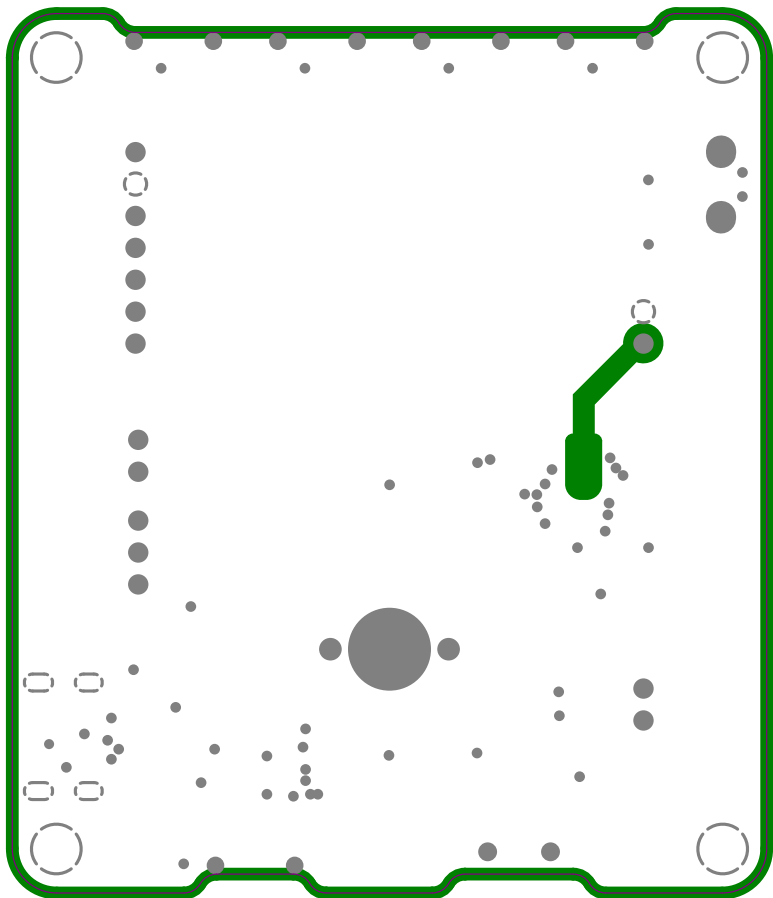
- Added cut-strap to allow users to limit detection threshold maximum

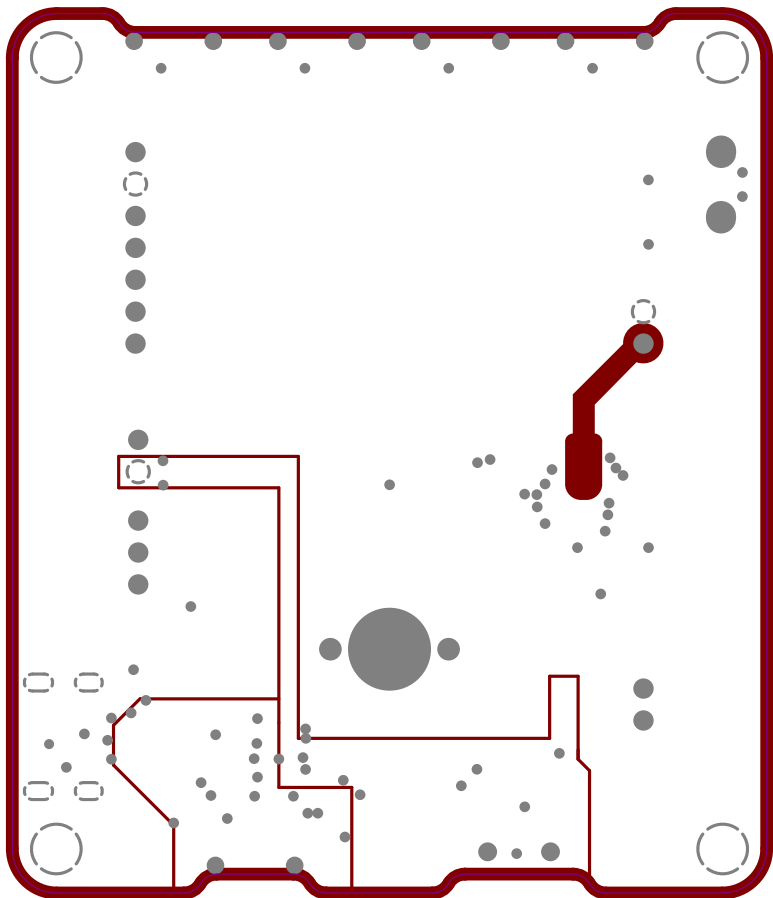
PCB:

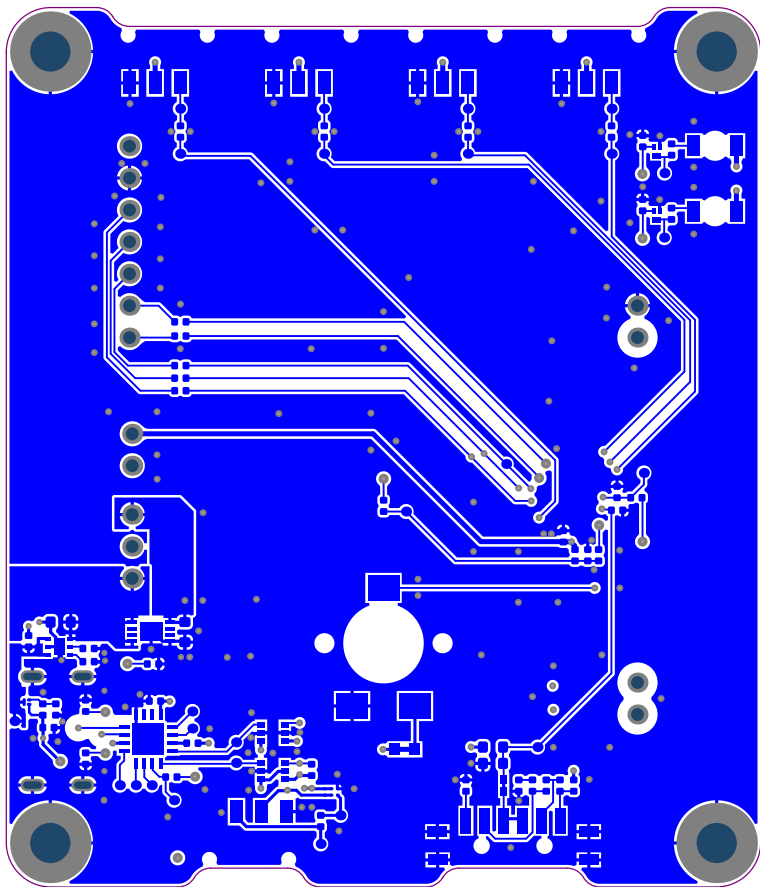
PCB revision 5

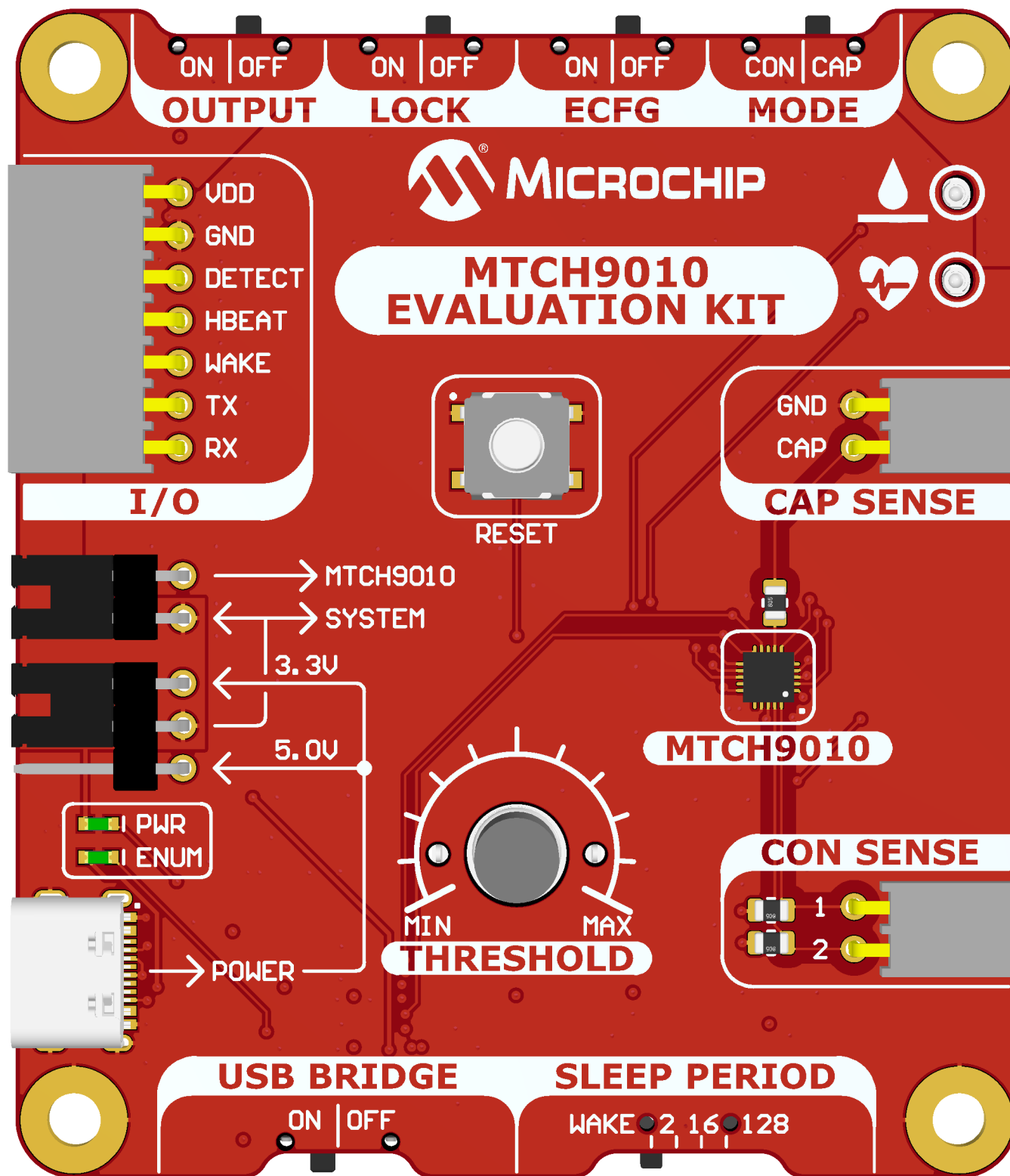
CNANO Template Revision: 2.8

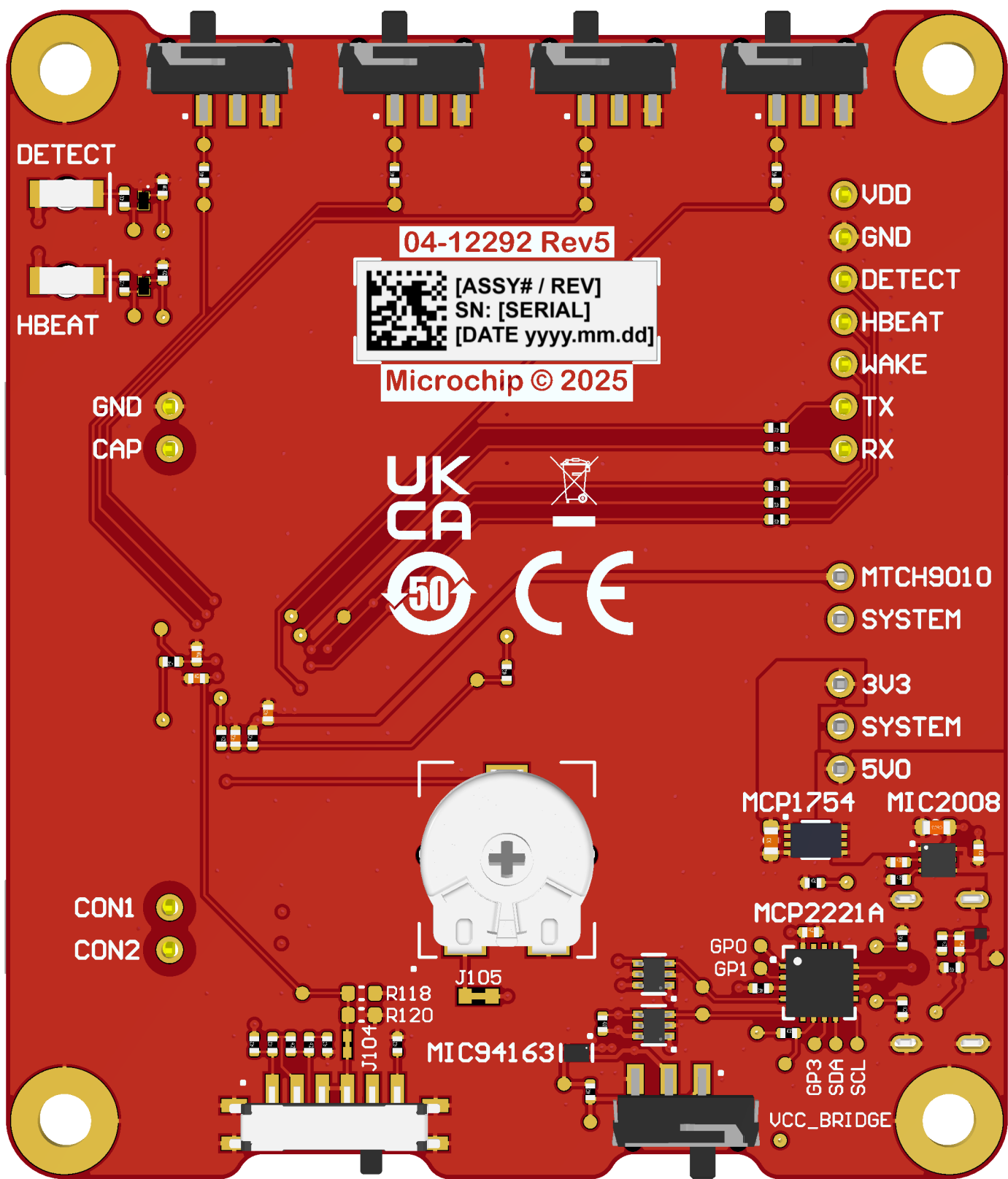


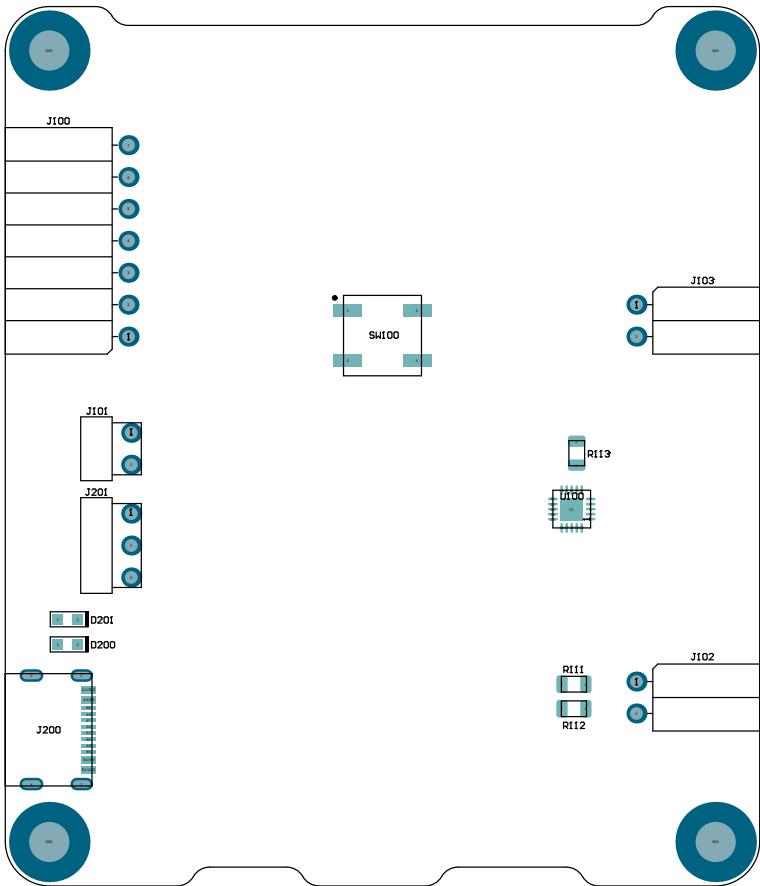


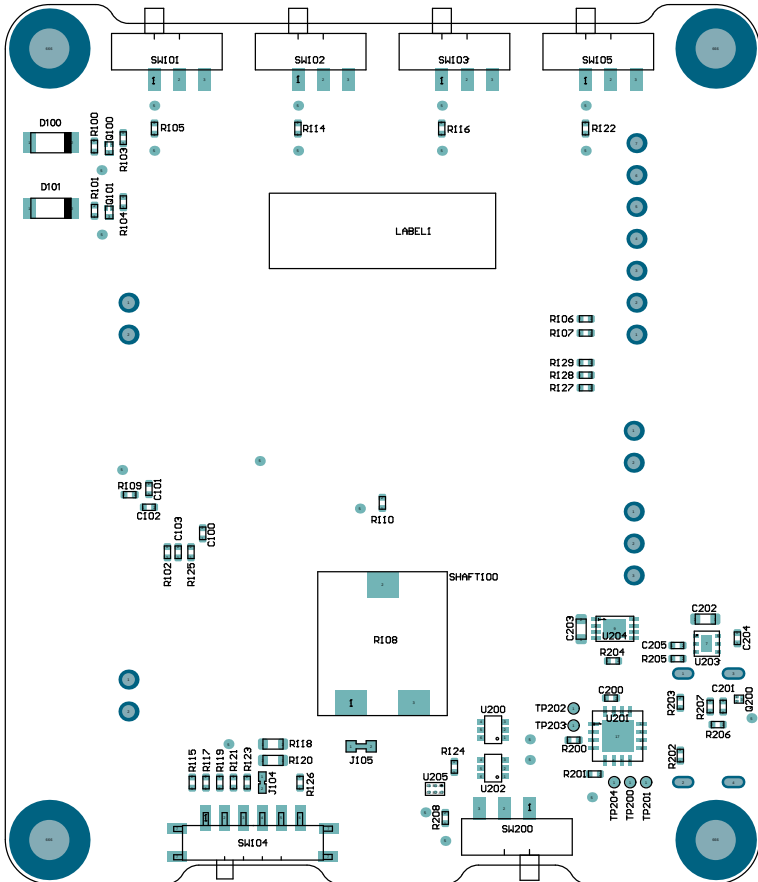




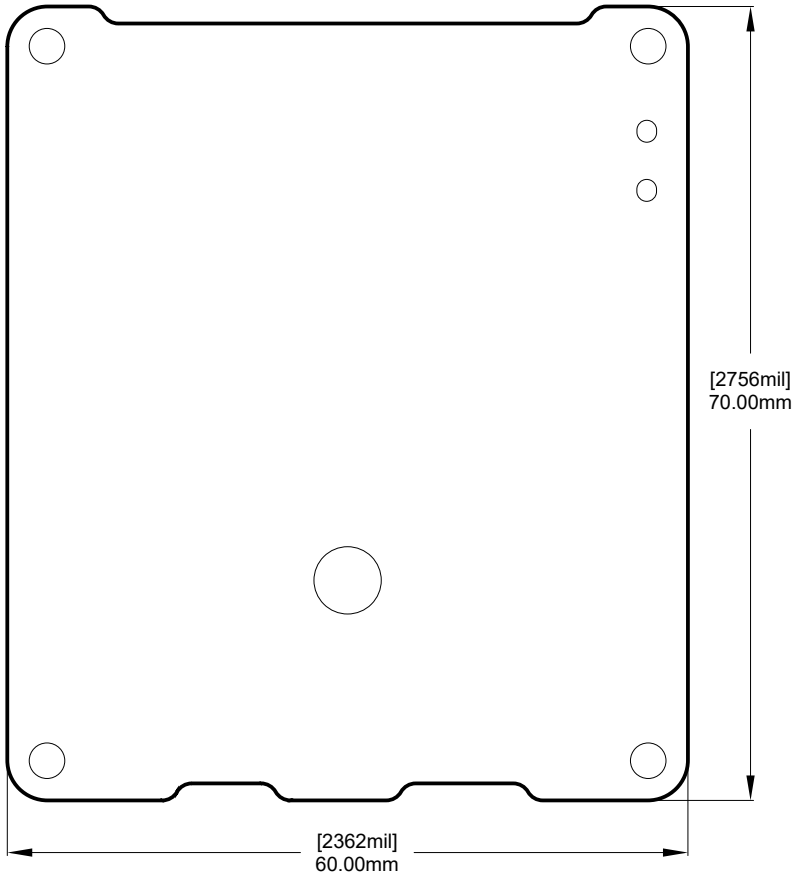




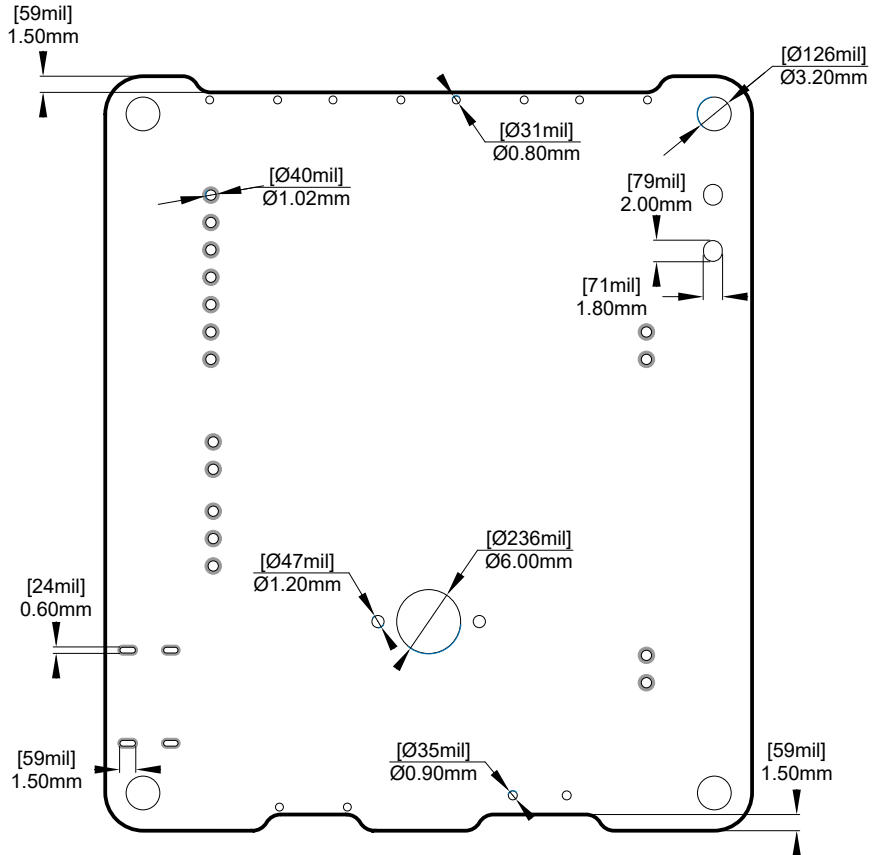




Board Outline



Through-Hole Dimensions



PCB Stackup

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.010mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.035mm		Signal	GTL
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Gnd Plane 1	0.035mm		Internal Plane	GP1
Core		1.200mm	FR-4	Dielectric	
Copper	Gnd Plane 2	0.035mm		Internal Plane	GP2
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Bottom Layer	0.035mm		Signal	GBL
Surface Material	Bottom Solder	0.010mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO
Total thickness: 1.594mm					

Component list

BOM for Variant [Default Assembly] of Project [MTCH9010_Evaluation_Kit.PrjPcb] (No PCB Document Selected)

Source Data From: MTCH9010_Evaluation_Kit.PrjPcb
Project: MTCH9010_Evaluation_Kit.PrjPcb
Variant: Default Assembly

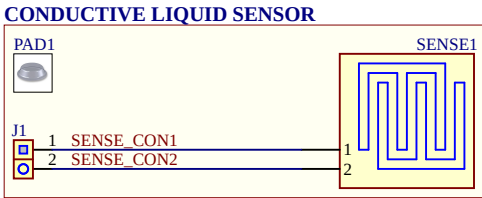
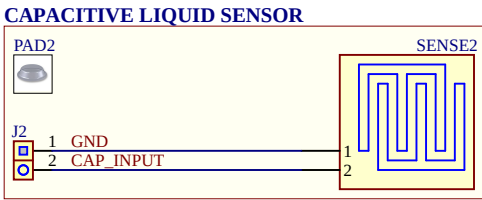


Report Date: 2025-04-14 14:13
Print Date: 2025-04-14 14:12:28

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Fitted	C100, C200, C201	3	0.1uF			
Fitted	C101, C102	2	1000pF			
Fitted	C103	1	10nF			
Fitted	C202, C203	2	4.7uF			
Fitted	C204	1	2.2uF			
Fitted	C205	1	2200pF			
Fitted	D100, D101	2	YELLOW			LED
Fitted	D200, D201	2	GREEN			LED
Fitted	J100	1	HDR-2.54 Female 1x7			
Fitted	J101	1	HDR-2.54 Male 1x2			
Fitted	J102, J103	2	HDR-2.54 Female 1x2			
Fitted	J200	1	USB2.0 Type-C FEMALE			
Fitted	J201	1	HDR-2.54 Male 1x3			
Fitted	JP100, JP200	2	JUMPER 2.54mm 1x2			
Fitted	LABEL1	1	PCBA LABEL 18X6mm			
Fitted	Q100, Q101	2	DMN65D8LFB-7B			Transistor MOS
Fitted	Q200	1	CSD25501F3			Transistor MOS
Fitted	R100, R101	2	2.2k			Resistor
Fitted	R102, R110, R205	3	470R			Resistor
Fitted	R103, R104, R117, R119	4	300k			Resistor
Fitted	R105, R106, R107, R109, R114, R116, R122, R126, R127, R128, R129, R200, R201, R204	14	1k			Resistor
Fitted	R108	1	100k			Resistor Var
Fitted	R111, R112, R113	3	10k			Resistor
Fitted	R115, R121, R123	3	100k			Resistor
Fitted	R124, R206, R207, R208	4	47k			Resistor
Fitted	R125	1	10k			Resistor
Fitted	R202, R203	2	5.1k			Resistor
Fitted	SCR1, SCR2, SCR3, SCR4	4	Phillips Screw 1/4"			
Fitted	SHAFT100	1	MISC PT10-SHAFT			
Fitted	STANDOFF1, STANDOFF2, STANDOFF3, STANDOFF4	4	1/4" Standoff Nylon			
Fitted	SW100	1	TACT SPST			Switch
Fitted	SW101, SW102, SW103, SW105, SW200	5	SLIDE SPDT			
Fitted	SW104	1	SLIDE SP4T			
Fitted	Test1	1				
Fitted	U100	1	MTCH9010			
Fitted	U200, U202	2	74LVC1T45DW-7			
Fitted	U201	1	MCP2221A			
Fitted	U203	1	MIC2008			
Fitted	U204	1	MCP1754/3.3V			
Fitted	U205	1	MIC94163YCS			
Not Fitted	R118, R120	0	0R			Resistor

Approved	87	Notes

MTCH9010 Sensors



LABEL1



[ASSY# / REV]
SN: [SERIAL]
[DATE yyyy.mm.dd]

PCBA LABEL 18X6mm

LABEL2



[ASSY# / REV]
SN: [SERIAL]
[DATE yyyy.mm.dd]

PCBA LABEL 18X6mm

"Text in Silkscreen on PCB"

S3
MTCH9010_Sensors_Revision_History.SchDoc

Project Owner: SLT		 MICROCHIP	
PCB Layout Contact: SLT			
PartNumber: EV24U22A		Project Title MTCH9010 Sensors	
Variant: Default Assembly			
Sheet Title MTCH9010 Sensors			
Size A3	SCH #: 02-01348 PCB #: 04-12409	Rev: 2 Date: 2025-04-14	Rev: 2 Sheet 1 of 2
File: MTCH9010_Sensors.SchDoc			

Designed with



Altium.com

Revision History

PCB Assembly Rev 1:

Design Changes:

Initial Design

PCB:

PCB revision 1

PCB Assembly Rev 2:

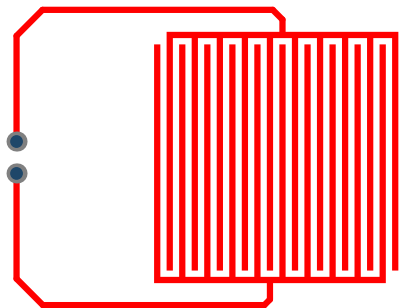
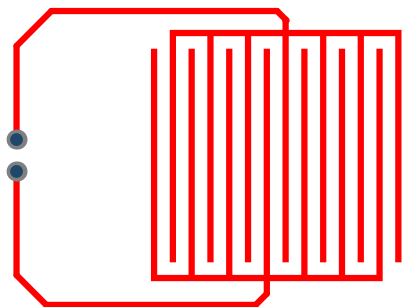
Design Changes:

- Capacitive sensor redesigned to reduce capacitance

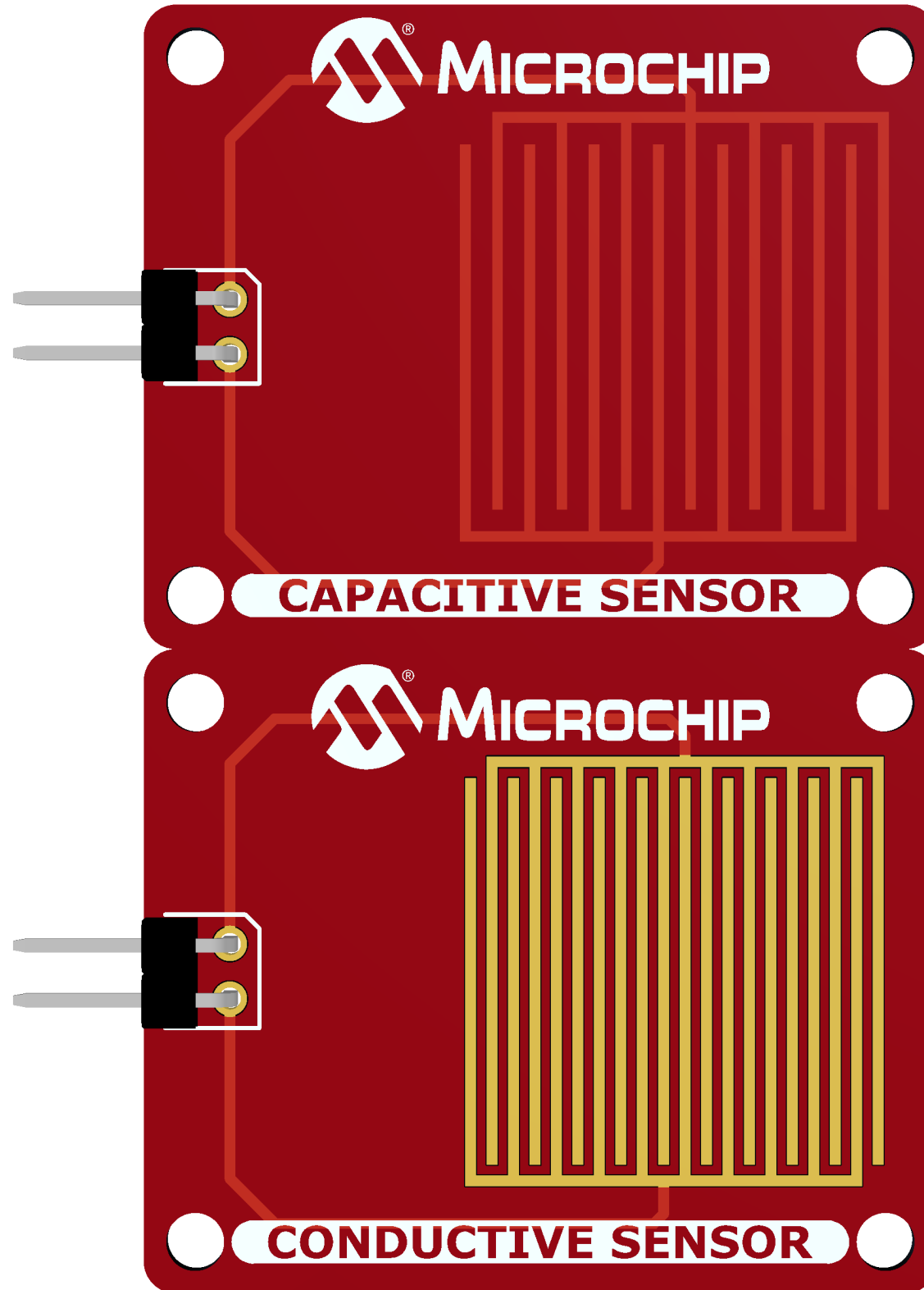
PCB:

PCB revision 2

CNANO Template Revision: 2.8









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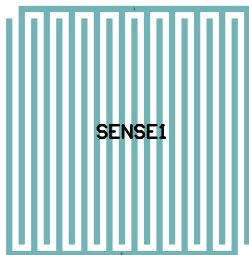
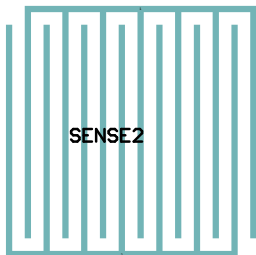
Microchip © 2025
04-12409B Rev2



[ASSY# / REV]
SN: [SERIAL]
[DATE yyyy.mm.dd]

Microchip © 2025
04-12409A Rev2





**Rubber
Pad**

LABEL2

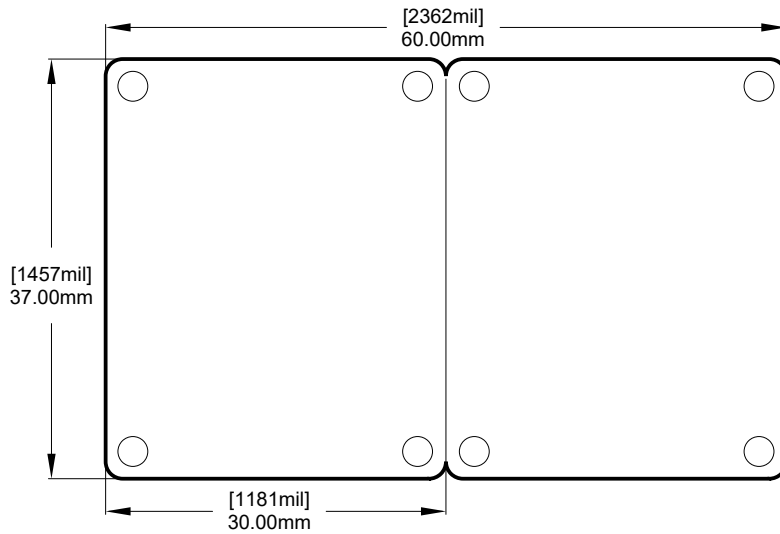


**Rubber
Pad**

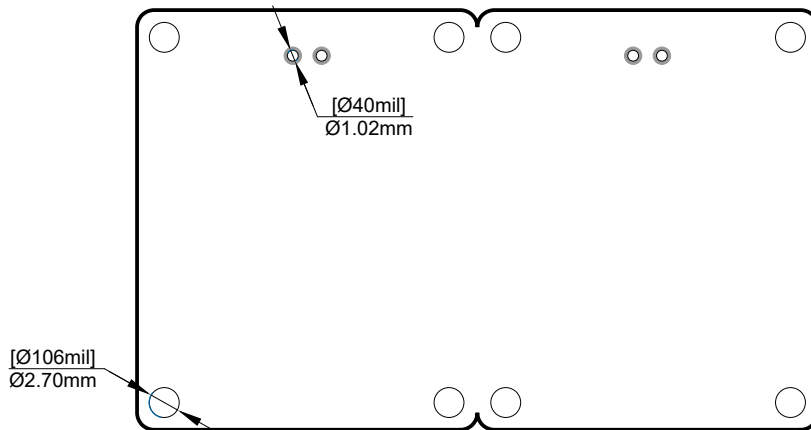
LABEL1



Board Outline



Through-Hole Dimensions



PCB Stackup

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.010mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.035mm		Signal	GTL
Core		1.500mm	FR-4	Dielectric	
Copper	Bottom Layer	0.035mm		Signal	GBL
Surface Material	Bottom Solder	0.010mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO
Total thickness: 1.590mm					

Component list

BOM for Variant [Default Assembly] of Project [MTCH9010_Sensors.PrjPcb] (No PCB Document Selected)

Source Data From:MTCH9010_Sensors.PrjPcb

Project:MTCH9010_Sensors.PrjPcb

Variant:Default Assembly



Report Date:2025-04-1414:22

Print Date:2025-04-1414:21:49

Fitted	Designator	Quantity	Value	#Column Name Error: Manufacturer	#Column Name Error: MPN	Description
Fitted	J1, J2	2	HDR-2.54 Male 1x2			
Fitted	LABEL1, LABEL2	2	PCBA LABEL 18X6mm			
Fitted	PAD1, PAD2	2	RUBBER PAD D12.7H6.35mm			

	6	
Approved	Notes	